Lab 3 Reflection

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Technical Description

I did all of Q3, which was applying PCA, t-SNE, and UMAP to the 1000 Genomes data. For 3A, I loaded the genotype data and the population labels. When I first tried to load the .npy file it wouldn't load, and I found out the file got cut down to basically nothing when it was moved into our 03_Lab folder, so I went back through our commit history and restored the original file. For 3B, the genotype data used -1 for missing values, so I used SimpleImputer to fill those with the column mean and then standardized everything with StandardScaler. For 3C and 3D, I reduced the data to 2 dimensions with PCA, t-SNE, and UMAP and made scatter plots colored by population. I set the random state to 1234 for all of them to match what my teammate used in Q2, which also fixed an issue where PCA was giving slightly different results every run. For 3E, I reduced the data to the top 25 PCA components first, ran t-SNE and UMAP again, and answered the reflection questions. UMAP gave the best separation, and doing PCA first made the groups even cleaner. I used Python libraries like numpy, pandas, sklearn, umap-learn, and seaborn for the code.

Positive Critique

Going back to Git again, this lab felt like the first time everything went smoothly on my end. When I started Q3 my branch was behind main since my teammate had already pushed Q2, so I updated my branch to main first before doing anything else. Then when I went to merge my work into main, another teammate had merged their Q1 answers in the meantime, but Git was able to combine everything without any conflicts. In Lab 1 I had to manually combine our code after overriding main, so being able to do all of this through Git with merges is exactly what I said I wanted to get better at. I also used our commit history again to restore the broken data file, which is the same thing that saved us in Lab 2. I'm happy that I'm actually getting more comfortable with Git each lab, and I think that's super important for working on a team.

Negative Critique

I want to not procrastinate next time, I couldn’t make it to the class the initial lab was assigned because of a conflict and then with fall break, I put off the lab for a little bit. But, when I got it the lab was completed fine, but for myself and my teammates it was better to just get it done earlier on and not have to worry about it moving forward especially going into midterm season.